Predicting Diabetes from BRFSS Health Indicators

A screening-focused machine learning project

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Project Question and Data

Question

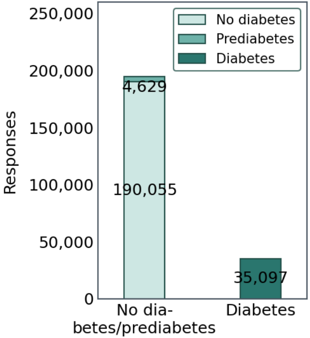
Can survey-style health indicators predict diabetes status well enough to support early screening?

- Source: CDC BRFSS 2015, via Kaggle
- Unit: one adult survey respondent
- Features: 21 health, lifestyle, and demographic indicators
- Intended use: follow-up prioritization, not clinical diagnosis

Cleaned Binary Dataset

229,781 rows

35,097 diabetes cases
15.3% positive class



Target and Cleaning Choices

Original Target

Label	Meaning
0	No diabetes
1	Prediabetes
2	Diabetes

Modeling Target

Label	Meaning
0	No diabetes or prediabetes
1	Diabetes

- Main pipeline removed 23,899 exact duplicate rows (9.42% of the raw data).
- A sensitivity check kept duplicates and reran the workflow.
- The duplicate choice changed some metrics, but did not change the main conclusion.

Modeling Strategy

Validation Design

- Stratified 80/20 train-test split
- 5-fold stratified cross-validation on training data
- Final metrics reported on the held-out test set

Models Compared

- Dummy baseline
- Logistic regression (regular, lasso, ridge)
- Random forest
- XGBoost
- Support vector machine
- MLP neural network

Primary KPI

Recall for the diabetes class

Secondary KPI

F2, AUPRC, AUROC, precision, F1, and TPR - FPR

Thresholds Drive the Screening Tradeoff

Threshold Strategies

Strategy	Role
Default	Standard cutoff
Max F2	Recall-heavy screening
Max TPR-FPR	Balanced ROC point

Lower threshold

- $\uparrow$ Recall
- $\downarrow$ Missed diabetes cases
- Better for screening

Higher threshold

- $\uparrow$ Precision
- $\downarrow$ False positives
- $\downarrow$ Workload and screening cost

Threshold choice is an operational decision, not only a modeling decision.

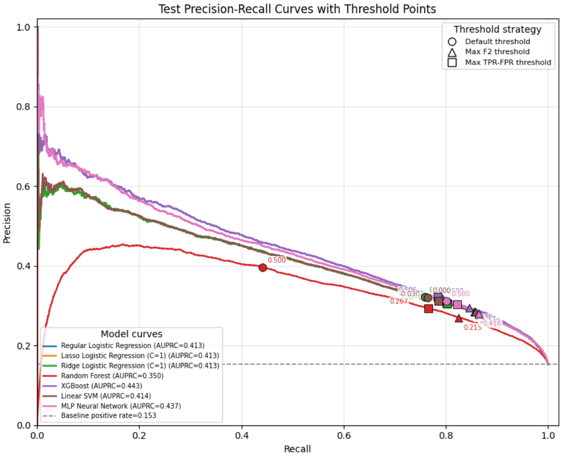
Top 5 Model-Threshold Choices by Test Recall

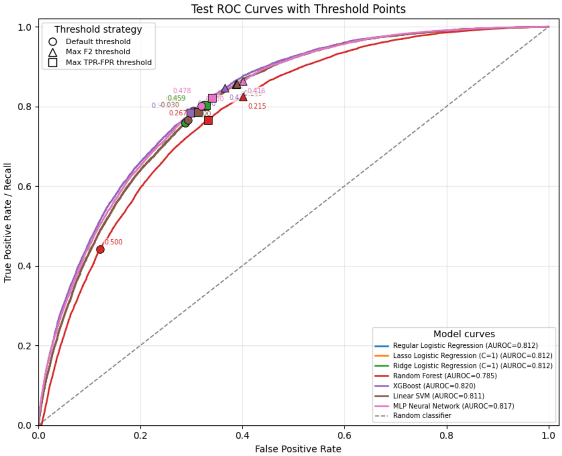
Highest Recall Operating Points

Model	Threshold Strategy	Threshold	Recall
MLP NN	Max F2	0.416	0.864
SVM	Max F2	-0.149	0.857
Ridge logistic ($C = 1$)	Max F2	0.402	0.856
Regular logistic	Max F2	0.402	0.856
Lasso logistic ($C = 1$)	Max F2	0.403	0.856

- Maximize the chances of catching every individual who actually has diabetes, minimizing dangerous false negatives
- Flag a massive number of healthy individuals as diabetic results in low precision, straining medical capacity and increasing costs

Test PR and ROC Curves





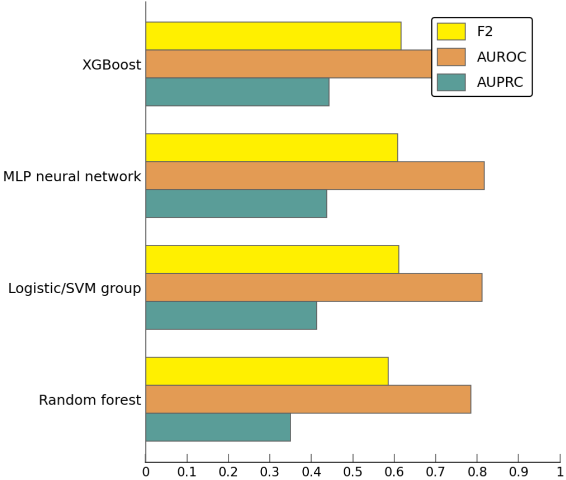
Curves and threshold markers are from the held-out test set in `notebooks/all_model_testing.ipynb`.

Beyond Recall: A Holistic View of Performance

Ranking Metrics

Model	AUPRC	AUROC	F2
XGBoost	0.443	0.820	0.616
MLP neural network	0.437	0.817	0.609
SVM	0.414	0.811	0.611
Regular logistic	0.413	0.812	0.611
Lasso logistic	0.413	0.812	0.611
Ridge logistic	0.413	0.812	0.611
Random forest	0.350	0.785	0.585

Baseline positive rate is 0.153, so AUPRC above 0.40 is meaningful.



Recommendation

Highest Recall	Best Ranking	Most Explainable
MLP	XGBoost	Logistic Regression
Use a recall-heavy threshold	Highest AUPRC and AUROC	Competitive and interpretable

- EDA and coefficients agree on plausible risk signals: general health, high blood pressure, BMI, age, high cholesterol, and difficulty walking.
- Practical deployment should choose a threshold based on follow-up capacity and false-negative cost.

Limitations and Next Steps

Limitations

- Self-reported survey data
- Predicts reported diabetes, not clinical diagnosis
- Prediabetes grouped with non-diabetes
- 2015 data may not fully reflect current populations
- Associations are not causal effects

Future Work

- Three-class modeling for prediabetes
- Cost- or capacity-based threshold selection
- Calibration and subgroup fairness checks
- External validation on newer BRFSS data